

Physical Layer Performance of DECT-2020 New Radio for Factory Automation

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Research and Innovation

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Abstract—Derived from the work at IMT-2020, the standard DECT2020-NR provides non-cellular communication solutions for applications requiring strict latency and reliability for flexible and adaptive deployments. Examples include industrial and audiovisual communications. This work-in-progress analyzes the physical layer performance of DECT2020-NR in an industrial environment. Different types of communication links are considered and various propagation channel conditions and system configurations are simulated. The results obtained confirm the need for a MAC access layer with link control techniques to ensure the reliability requirements of industrial applications.

Index Terms—DECT-2020NR, Factory Automation, URLLC, mMTC, 5G, Physical layer

I. INTRODUCTION

The fourth industrial revolution and its requirements impose radical changes to the underlying networking technologies. Drath et al. [1] state that the major technical background of Industry 4.0 is the introduction of Internet technologies into industry. Thus, the performance of radio communication systems in automation applications is challenging. In particular, in industrial environments, with many metal surfaces and moving objects, are significantly demanding for wireless transmissions. The large fluctuation in the quality of wireless transmission channels compared to wired ones (e.g., high end-to-end latency and reduced reliability) raises the necessity to address new technologies and techniques to enhance and ensure the required-by-industry network quality.

In addition to this new paradigm, Internet of Things (IoT) applied to the industry was transformed into Industrial Internet of Things (IIoT). IIoT represents the use of IoT in manufacturing [2], [3]. It can be defined as the development of connected devices that provide sensing and processing capability. Currently, on the factory area, wired connections provide highly reliable but costly and physically-restrictive connectivity. Wireless connectivity, however, offers many benefits over wired solutions. First, lower expenditures and decreased long-term maintenance costs can be achieved with wireless solutions by the elimination of conduit and cable. Second, wireless connections can be used in otherwise impractical locations, using low-power monitoring devices, thus eliminating the difficulties inherent in physically routing cables. Third, wireless communication allows quick mobility and

reconfigurability; with wireless communication, it is possible to have easily relocatable and reconfigurable workcells [4]. It is because of this that it has been decided to study an IoT technology that allows it to be used in harsh scenarios such as industry. In particular, the new DECT-2020NR standard has been established as an option to be studied in this paper.

The DECT-2020NR standard has been designed to meet the requirements of URLLC (Ultra-Reliable Low Latency Communication) and mMTC (massive Machine Type Communications), which are prevalent in industrial environments. However, there is a scarcity of information and content regarding this new standard. Hence, there exists a need to model DECT-2020NR at a system level. As a first step towards achieving these models, it is necessary to create physical layer models that fit these industrial communication scenarios. Several works already exist where DECT-2020NR is modeled [5], [6]; however, the use cases addressed in them do not represent industrial scenarios. Therefore, the purpose of this work is to fill this gap and generate suitable physical layer models that adapt to these industrial scenarios, which are parameterized according to the literature. These industrial scenarios are based on proprietary use cases.

This paper studies and evaluates the physical layer of the DECT-2020NR standard for a factory automation environment. First, the main characteristics of DECT-2020NR are presented. In Section II, the defined use case for this environments is detailed. This use case includes two main types of scenarios: URLLC; and mMTC. The study that has been carried out on each of the scenarios is presented, including their particular characteristics. In Section III, the associated physical layer simulations with two of those analyzed scenarios are presented. The next section discusses the obtained results, including the behaviour of the Bit Error Rate (BER) and Packet Error (PER). Finally, the obtained conclusions are presented as well as the future work and lines that will be performed.

A. DECT-2020NR standard

DECT-2020NR is a Radio Interface Technology standard which is part of the 5G paradigm. It emerged within IMT-2020 as a solution to the URLLC and mMTC scenarios.

Although it is a technology which is able to operate in the rest of the 19 bands defined in IMT-2020, it has a dedicated frequency band (1880-1900 MHz). It has Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) to manage radio resources, Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA) as media access methods. It defines 4 main layers: physical layer, Medium Access Layer (MAC), Data Link Control (DLC) and Convergence (CVG).

Physical layer can use different bandwidths in the channels for transmission: 1,728, 3,456 and 6,912 MHz. It presents 12 types of modulation and coding schemes, which range from the most robust (BPSK, code rate of 1/2), to the scheme that offers the greatest capacity (1024 QAM, code rate of 5/6). The physical layer also offers different numerologies that allow adjusting the spacing between subcarriers as well as the scale factor of the Fourier transform. The packets are divided into 4 parts: the STF (Synchronization Training Field) synchronization signal, the DRS (Demodulation Reference Signal) demodulation, the control channel and finally the data channel. This physical layer uses frames of 10 ms duration that are divided into 24 slots. At the same time, each slot can be divided into subslots, depending on the used numerologies [7].

The MAC layer is one of those that gives the greatest differences to the standard. This layer allows radio devices (RD) to operate in fixed (FT) or portable termination (PT) modes or both at the same time. A device in FT mode works as router coordinating local radio resources and coordinating network communications. PT mode is very similar to the role that user equipments (UE) assume in 5G networks, where they manage their own communications and connect to a neighbouring FT node. This layer allows devices to select the mode to operate (FT, PT or both) according to the state of the network. MAC layer provides 3 channels to transmit: broadcast channel (BCH/PCH); random access channel (RACH); and a dedicated channel (DCH). The usage of deterministic communications through the DCH channel is also possible. DCH allows scheduled access through the reservation of radio resources by the FT node for a given communications flow. Furthermore, it also allows random transmissions using RACH and listen before talk (LBT) algorithms for access to the medium [8].

Finally, DLC and CVG layers are proprietary to the standard. The DLC layer manages each of the radio links of RDs, as well as the routing in these DECT networks. CVG layer establish transmission procedures based on traffic characteristics and provides compatibility with other protocols outside the scope of this standard [9].

II. USE CASE

The DECT-2020NR standard was designed for these two main scenarios. The presented use case encompasses both, which are focused on achieving reliability with low latency and scalability in an automation process context. Concretely, this

use case describes a complete industrial environment of a production automation factory. In this environment five different scenarios have been identified based in their communications requirements.

The first one is the production line scenario, here several robotic arms collaborate in a small area to assemble multiple parts of a product. This robotics arm have to be coordinated to work together, so the synchronization of them is critical. This makes the scenario a URLLC type scenario, where the requirements of reliability and latency are very strict. The communications requirements of this scenarios have been decided according to [3]

Then we have the security cameras scenario, where numerous cameras are strategically positioned throughout the factory, with a particular focus on areas near the production line. These cameras are deployed to oversee and ensure the security of the factory premises. The communication requirements in this domain are aligned with those specified in [10] for security camera systems

Related with security we have another scenario, this is the safety one. There, sensors are strategically placed around the production line workcell with the objective of responding to other existing sensors integrated into operators' helmets. If an individual breaches the perimeter without adhering to proper safety protocols, the sensors surrounding the workcell will trigger a response, and the robotic arms must stop. These scenario have very strict safety requirements as established in [3].

Around the factory several autonomous vehicles are deployed, this vehicles bring materials from the warehouse and they also carry the products from the production line to the storage room. They communicate between each other and with the concentrator node. The location data is sent to the concentrator and the corresponding instructions and actions to perform are received from it. The vehicles communicate between them to avoid collisions. Given that the maximum working speed of this vehicles is low (few m/s), the communication requirements are not strict in terms of latency. Although they send a large amount of information per minute to the network, these communications meet the mMTC profile.

The last scenario is the monitoring one, this scenario corresponds to all the sensors distributed throughout the factory that collect information for the monitoring and analysis of all the processes that take place in the factory. The sensors are mainly distributed in the production line and in the autonomous vehicles. They communicate with the concentrator nodes which collect this information and send it for further analysis. Due to the large number of equipment involved in this monitoring this scenario represents the mMTC characteristics, its requirements are based on those analysed in [10]

These previous casuistics provides 5 main communications scenarios: 1) the production line which includes communications between robots; 2) safety sensors for operators' safety; 3) security cameras to receive information from the cameras; 4) autonomous vehicle control communications to avoid collisions and orchestrate their operation; and 5) monitoring,

where both the workcell and autonomous vehicles sensors are considered. The requirements for each one are depicted below, where we define the latency, PER and the amount of data the physical layer have to send.

TABLE I: Requirements for each Use Case scenario

N°	Scenarios	Latency (ms)	Reliability (PER)	Data (bits)
1	Production Line	1	10^{-7}	136
2	Safety Sensors	1	10^{-8}	136
3	Security Cameras	100	10^{-3}	50 M
4	Vehicle Control	10	10^{-4}	10 M
5	Monitoring	100	10^{-3}	512

III. SIMULATION ENVIRONMENTS

This section introduces the simulation environment employed to simulate the scenarios outlined in the preceding section. Specifically, simulations for two out of the five defined scenarios, namely the Production Line and Security Cameras ones, are presented herein. These simulations were conducted at the physical layer of DECT-2020NR using the MATLAB simulation program. The simulation code utilized in this study is derived from the work of Maxim Penner, as documented in [5]. The following presents the specific parameters with which the scenarios have been simulated in MATLAB, as well as the links associated with each of the scenarios.

A. Physical Layer parametrization

For both scenarios, a perfect channel estimation and synchronization have been assumed on the receiver. The physical layer header sent is 40 bits long for both scenarios, because we want to maximize the transmission of useful information. The channels, as well as the transmission frequency, bandwidths and other parameters, are specific to each one.

For the production line scenario, communications need to be as robust as possible, which is why the physical layer numerology used is the one seen in Table II.

For the security cameras, physical layer parameters have been chosen to achieve the specified throughput of 50 Mbps. Hence, a less robust numerology has been employed, and the transmission bandwidth has been increased. The transmission frequency differs from that used in the production line case to avoid interference with these scenario communications.

TABLE II: Physical layer simulation parameters for Production Line and Security camera scenarios

Parameters	Scenarios	
	Production Line	Security Cameras
Modulation	BPSK	16 QAM
Code Rate	1/2	3/4
μ, β	1, 1	4, 4
Bandwidth	1.728 MHz	6.912 MHz
Transmission mode	SISO	SISO
Frequency	1.4 GHz	0.9 GHz

B. Channel parameters

After configuring the physical layer of the devices, a study of the radio links generated for this scenarios has been carried out. It has been observed that two different radio links are generated, one for each scenario. These links have been modeled with specific channels that allow them to be precisely parameterized. The characteristics of the links of each scenario can be seen in Table III.

TABLE III: Simulated channels for Production Line and Security camera scenarios

	Delay Spread	LoS/NLoS	Clusters	Taps	Doppler
Production Line	30 ns	LoS	8	32	15 Hz
Security cameras	100 ns	LoS	4	12	0 Hz

The production line link is characterized by a high number of metallic surfaces surrounding the cell, leading to numerous multipath components. This is reflected in the considerable number of echoes (taps) and clusters observed. Due to the small cell area, the delay spread is approximately 30 ns, satisfying the condition outlined in [11]. This condition stipulates that a delay spread of less than 60 ns is necessary for the proper functioning of industrial communications of this kind. The transmitters and receivers are arranged in such a way that there is almost always a line of sight between them. Therefore, a Line-of-Sight (LoS) channel has been assumed for simulation purposes. Finally, the movement of the robotic arms results in a maximum Doppler shift of 15 Hz. Therefore, this link has been modeled using the channel described in [12] for the mechanical room scenario.

The links between the cameras and the concentrator have a greater distance than that of the production line, resulting in a higher observed delay spread, approximately around 100 ns. Additionally, due to being in an area with fewer obstacles, the received echoes are much less pronounced and are more widely spaced over time. The channel formed between the devices is LoS because the equipment is positioned on the ceiling, ensuring a constant line of sight between them. Due to these characteristics and considering the data in Table III, the WLAN 802.11ah channel, model D [13], has been chosen as the basis for simulating the links in these scenarios.

IV. RESULTS AND DISCUSSION

This section presents the results obtained by simulating the physical layer of the scenarios defined in section III. The evolution of the BER and PER for the two mentioned scenarios is presented. These SNR/PER curves have been simulated for a range of values from -10 to 20 dB Signal to Noise Ratio (SNR). A total of 10 million packets have been simulated for each simulated SNR.

In the production line scenario, transmitted packets have been of a size of 136 bits for the data channel, while for the security camera scenario, packets of 23,048 bits have been sent. Below we presents the results obtained with the numerologies described in the previous section, which are the ones that provide the highest performance for each scenario.

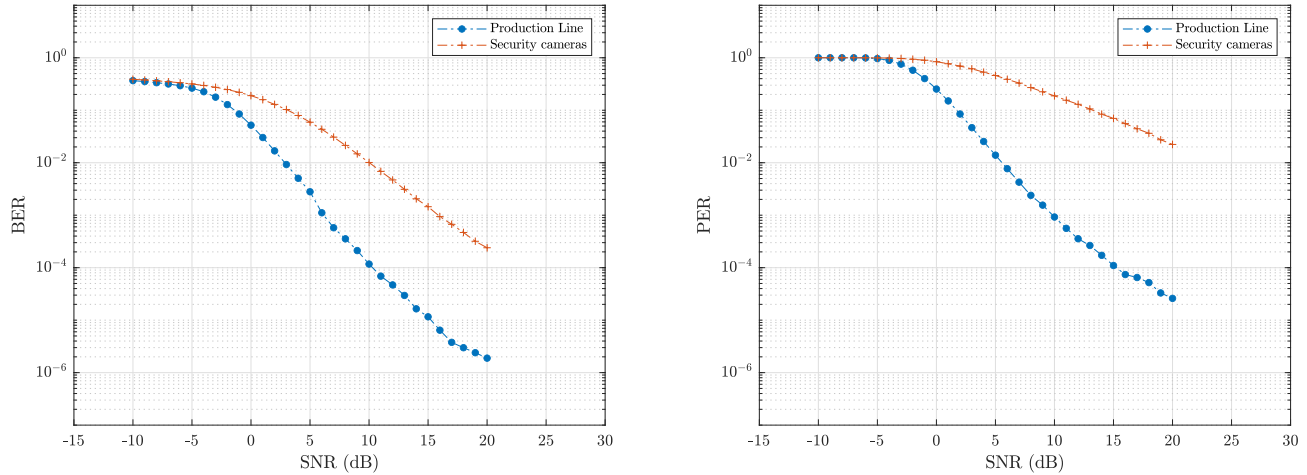


Fig. 1: BER and PER results for the simulated scenario

These simulations demonstrate that the physical layer alone is insufficient to meet the specific requirements of each scenario. As observed in Figure 1, we do not achieve the required 10^{-7} threshold for production line communications; the minimum PER remains at 10^{-5} . For the security cameras, the simulations yield similar results, with a minimum PER close to 10^{-2} . Therefore, the implementation of upper-layer protocols becomes necessary to address these limitations.

V. CONCLUSIONS AND FUTURE WORK

The main scenarios based on wireless communications which can be deployed in an industrial facility have been defined and studied. For these ones, the communication capabilities provided by DECT-2020NR technology have been analysed from the physical layer perspective. The results of the simulations of the most representative scenarios have been analysed and presented, using the optimal numerologies for each of them. In the future it is intended to complete the work simulating all the scenarios addressed in this paper.

This physical layer is not enough to meet the needed communication requirements in these types of environments. Thus, it is intended to implement all the defined layers of the standard. These layers can implement repetition, routing and positioning algorithms which can improve the results obtained by using only the physical layer. Furthermore, other relevant but not verified (in this paper) metrics can be validated (e.g. latency). To do so, the defined use case is going to be implemented and generated in an event simulation software (e.g. NS-3). The results presented in this paper will serve as the foundation for the event simulation implementation, wherein the PER/SNR curves simulated using this Matlab code will represent the physical layer in the simulator.

Finally, it is intended to complete the research on this technology by designing Self-Organizing Network (SON) algorithms. They can allow the exploitation of the reconfigura-

tion, adaptability, and scalability capabilities that the DECT-2020NR standard offers.

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